Lecture 4 recap

In Lecture 4, we:

- Learnt that an nth-order linear ODE is equivalent to a system of n 1st-order linear ODEs
- In general, treat vector-valued, linear ODEs of form

$$\boldsymbol{y}' = \mathbf{A}\,\boldsymbol{y}$$

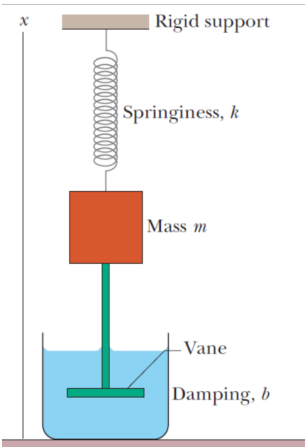
- Discovered a connection between the solution of the homogeneous, constant-coefficient problems with eigenvalues and eigenvectors of a matrix.

Lecture 5

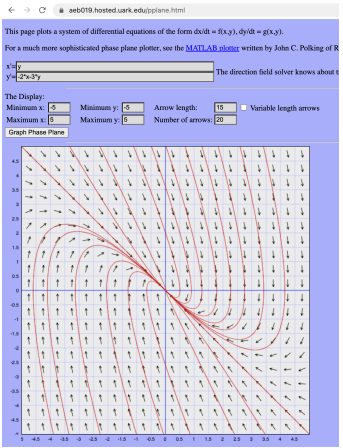
- In Lecture 5, we will:
  1. Learn how to visualise solutions to 2×2 homogeneous systems as trajectories in phase space
  2. Begin understanding how to sketch phase portraits by including critical points and straight line trajectories

How does linear algebra dictate structure of solution space?

Physical space



Abstract phase space



Right-hand figure is from a webpage where you can plot phase portraits

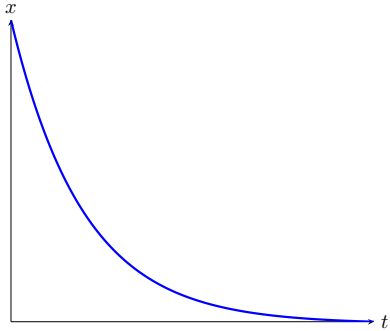
2nd order constant coefficient systems

The mass–spring system is a canonical example of a 2nd-order constant coefficient ODE

$$m\,x'' + b\,x' + k\,x = 0 \quad \text{where} \quad m, b, k > 0$$

Overdamped case

$$x(t) = C_1\,\mathrm{e}^{-A\,t} + C_2\,\mathrm{e}^{-B\,t}$$



We examine the overdamped case and leave the other cases for self-directed learning.

Exercise 1.12: Which case is the system from Problem 1.2?

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 3\frac{\mathrm{d}x}{\mathrm{d}t} + 2x = 0$$

or $\quad \boldsymbol{y}' = \mathbf{A}\boldsymbol{y}$ $\quad$ where $\quad \boldsymbol{y} = \begin{pmatrix} x \\ v \end{pmatrix}$ $\quad$ and $\quad \mathbf{A} = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$

$$\begin{pmatrix} x \\ v \end{pmatrix} = C_1\mathrm{e}^{-t}\begin{pmatrix} -1 \\ 1 \end{pmatrix} + C_2\mathrm{e}^{-2t}\begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

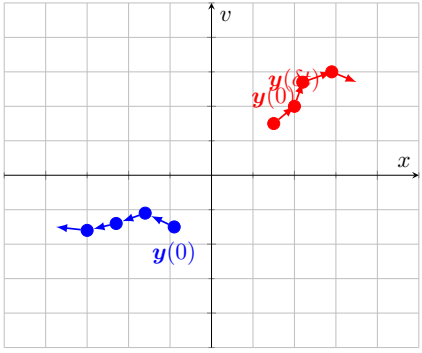
Real and distinct eigenvalues ($\lambda = -1$ and -2) = overdamped

- Fix C_1 and C_2 with ICs $x(0) = x_0$ and $x'(0) \equiv v(0) = v_0$ (or $\boldsymbol{y}(0) = \boldsymbol{y}_0$)
- Let's leave them unspecified for now.

We construct the phase portrait to establish the structure of solution space.

Phase space

- The xv-plane is called the **phase space***
- Seek solution as a planar parametric curve in t, called the **trajectory** or **orbit**
- Consider phase point $\boldsymbol{y}(0)$, representing the state of the system at time $t = 0$.
- $\boldsymbol{y}'(0) = \mathbf{A}\boldsymbol{y}(0)$ points along the trajectory
- Trace out trajectory by repeating this process
- Different $\boldsymbol{y}(0)$ leads to different trajectory
- Uniqueness of IVPs implies that trajectories cannot cross.



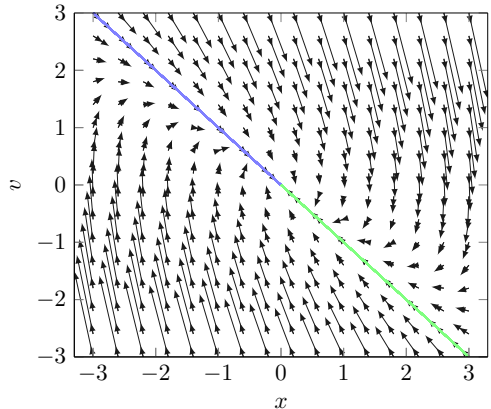
* In physics, phase space is frequently defined in terms of position and momentum instead of position and velocity

Vector field

- Plot vectors $\boldsymbol{y}'$ at mesh of points in phase space
- For Problem 1.2, plot

$$\begin{pmatrix} v \\ -2\,x - 3\,v \end{pmatrix}$$

  for different points (x, v) in phase space.
- Can help visualisation to normalise vector lengths, i.e., make vectors all same length



Critical points (or stationary points or equilibria)

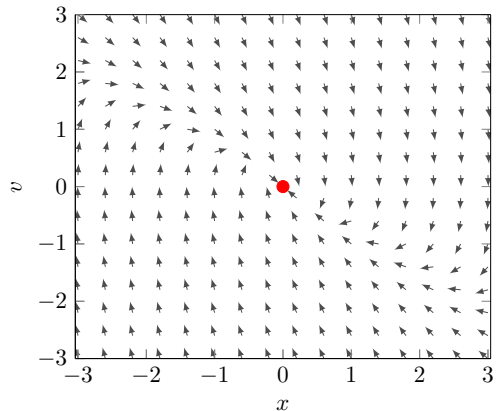
- Phase points $\boldsymbol{y}(t)$ such that $\mathbf{A}\boldsymbol{y} = \mathbf{0}$ are **critical points** of $\boldsymbol{y}' = \mathbf{A}\boldsymbol{y}$.
- At critical points $\boldsymbol{y}' = \mathbf{A}\boldsymbol{y} = \mathbf{0}$, and thus the solution does not change in time.
- In our example (Problem 1.2), we find $\boldsymbol{y}(t)$ such that

$$\underbrace{\begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}}_{\mathbf{A}} \boldsymbol{y} = \mathbf{0} \quad \Rightarrow \quad \mathbf{y = 0}$$

- Only critical point of linear system $\boldsymbol{y}' = \mathbf{A}\boldsymbol{y}$ for $\mathbf{A}$ invertible is $\boldsymbol{y} = \mathbf{0}$

Phase portraits

- *Phase portrait* is a graphical representation of the solution space
- Contains all critical points
- Contains representative orbits of system
- Matrix **A** alone shapes the structure of the solution space



Sketching phase portraits: Straight line orbits

What happens if IC is eigenvector of $\mathbf{A}$?

- Then $\mathbf{A}\boldsymbol{y}_0 = \lambda \boldsymbol{y}_0$
- So $\boldsymbol{y}'$ maintains its direction
  - Except possibly for a flip
- Orbit is straight line pointing in direction of increasing t
- For Problem 1.2
  - One eigenvector is $(-1, 1)^T$
  - Another is $(-1, 2)^T$

